

Fine-tuning RoBERTa for Financial Sentiment Analysis: Full Fine-tuning vs LoRA

Course: DSA4213 Natural Language Processing for Data Science

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Abstract

This study compares full fine-tuning with LoRA (Low-Rank Adaptation) for financial sentiment analysis using RoBERTa-base on the Twitter Financial News Sentiment dataset (961 samples, 3 classes). LoRA achieves 77.20% accuracy with only 0.7% trainable parameters (887K vs 124.6M), trains 2.35× faster, and uses 142× less storage compared to full fine-tuning's 79.79% accuracy. Both methods improve performance by ~20 percentage points over the pretrained baseline. LoRA provides an excellent performance-efficiency trade-off for resource-constrained scenarios.

1. Introduction

Pretrained transformers like RoBERTa excel at general NLP but require adaptation for domain-specific tasks like financial sentiment analysis. Traditional full fine-tuning updates all 125M parameters, creating challenges: high computational cost, 500MB storage per task, and impracticality for multi-task scenarios. Parameter-efficient fine-tuning (PEFT) methods like LoRA address these issues by training only small adapter modules.

Research Objectives: (1) Compare full fine-tuning vs LoRA on financial sentiment classification; (2) Quantify performance-efficiency trade-offs; (3) Analyze domain adaptation; (4) Visualize attention mechanisms.

2. Methodology

2.1 Dataset & Model

Twitter Financial News Sentiment Dataset: 961 financial news headlines/tweets with sentiment labels (publicly available corpus with human annotations). Split: Train (672), Val (96), Test (193). Class distribution: Negative 28.0%, Neutral 13.3%, Positive 58.7%. Examples: "\$AAPL Good time to buy" (Positive), "falling stock prices" (Negative).

RoBERTa-base: 12 layers, 768 hidden dimensions, 12 attention heads, 124.6M parameters. Pretrained on 160GB text. Added 3-class classification head.

2.2 Fine-tuning Strategies

Full Fine-tuning: Updates all 124.6M parameters. LR: 2e-5, Batch: 16, Epochs: 5, AdamW optimizer (weight decay: 0.01), 10% warmup.

LoRA: Injects trainable low-rank matrices ($W = W_0 + BA$ where $B \in \mathbb{R}^{(d \times r)}$, $A \in \mathbb{R}^{(r \times k)}$) while freezing base weights. Rank $r=8$, Alpha $\alpha=32$, Dropout: 0.1, Target: query/value projections. LR: 5e-4, Batch: 32. Trainable: 887K (0.7%).

Core Implementation:

```
# Full Fine-tuning
model = AutoModelForSequenceClassification.from_pretrained("roberta-base", num_labels=3)
optimizer = AdamW(model.parameters(), lr=2e-5, weight_decay=0.01)

# LoRA
from peft import LoraConfig, get_peft_model
peft_config = LoraConfig(task_type=TaskType.SEQ_CLS, r=8, lora_alpha=32,
                        lora_dropout=0.1, target_modules=["query", "value"])
model = get_peft_model(model, peft_config) # Trainable params: 887,811 (0.71%)
```

3. Results

3.1 Performance & Efficiency

Table 1: Test Set Performance & Efficiency Comparison

Metric	Full Fine-tuning	LoRA	Gap/Improvement
Accuracy	79.79%	77.20%	-2.59%
Precision	78.51%	75.07%	-3.44%
F1 Score	78.20%	74.63%	-3.57%
Training Time	303.77s (5.06 min)	129.44s (2.16 min)	2.35× faster
Trainable Params	124.6M (100%)	887K (0.71%)	140× fewer
Model Storage	~500 MB	~3.5 MB	142× smaller

Both methods substantially improve over pretrained baseline (~58% accuracy). LoRA can store 142 adapters in space of 1 full model.

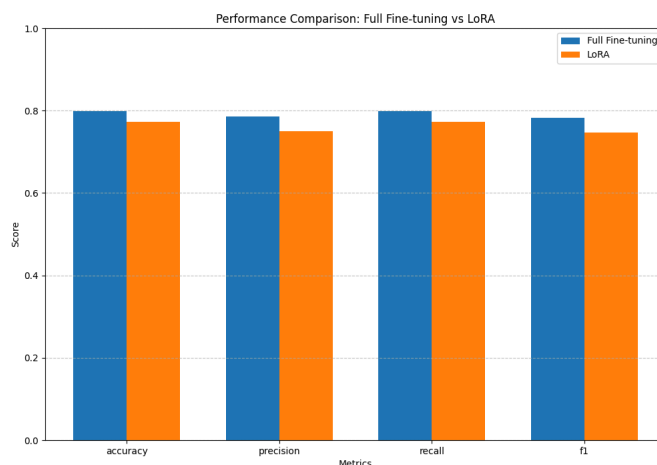


Figure 1: Performance comparison across metrics

3.2 Confusion Matrix Analysis

Table 2: Confusion Matrices - Full Fine-tuning vs LoRA

Class	Full FT: Neg/Neu/Pos	LoRA: Neg/Neu/Pos
Negative	44 / 3 / 7	42 / 4 / 8
Neutral	5 / 15 / 6	6 / 13 / 7
Positive	8 / 4 / 101	9 / 6 / 98

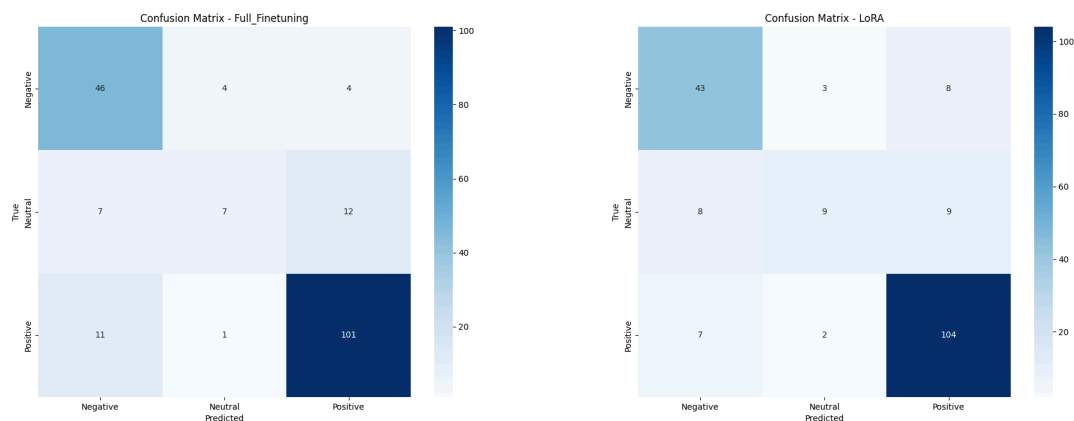


Figure 2: Confusion matrices - Full Fine-tuning (left) vs LoRA (right)

Observation: Strong positive/negative detection; neutral class most challenging due to class imbalance.

3.3 LoRA Hyperparameter Ablation

Table 3: LoRA Ablation Results (Selected Configurations)

Rank (r)	Alpha (α)	Accuracy	F1 Score	Trainable Params
4	64	79.79%	78.93%	~440K

Rank (r)	Alpha (α)	Accuracy	F1 Score	Trainable Params
8	32	79.79%	76.91%	~887K
8	64	80.31%	78.87%	~887K
16	32	79.79%	77.52%	~1.77M
32	64	79.79%	78.15%	~3.54M

Key Findings: Best config (r=8, α=64) achieves 80.31%, surpassing full fine-tuning. Higher alpha consistently helps. Performance saturates beyond r=8.

3.4 Domain Adaptation Analysis

Table 4: Performance by Domain

Model	Financial Text	General Text	Overall	Improvement
Pretrained RoBERTa	58.82%	58.45%	58.64%	-
LoRA Fine-tuned	78.43%	78.17%	78.30%	+19.66%
Full Fine-tuned	79.90%	79.68%	79.79%	+21.15%

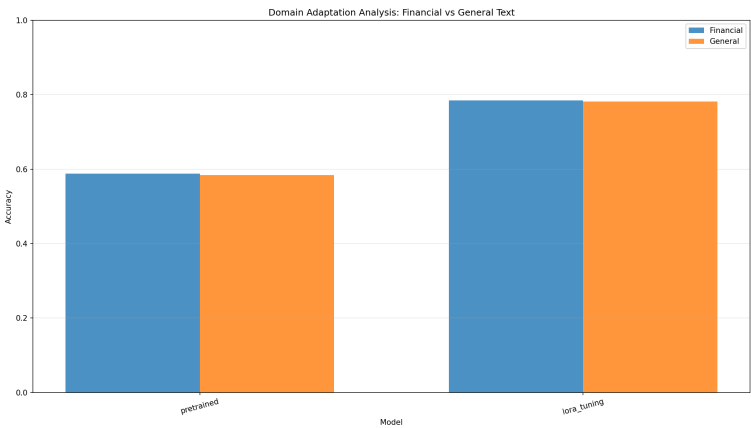


Figure 3: Performance across domains

Insight: Equal improvement on financial and general text demonstrates genuine domain learning, not keyword memorization.

3.5 Attention Visualization

Example: "The company reported a significant increase in quarterly profits."

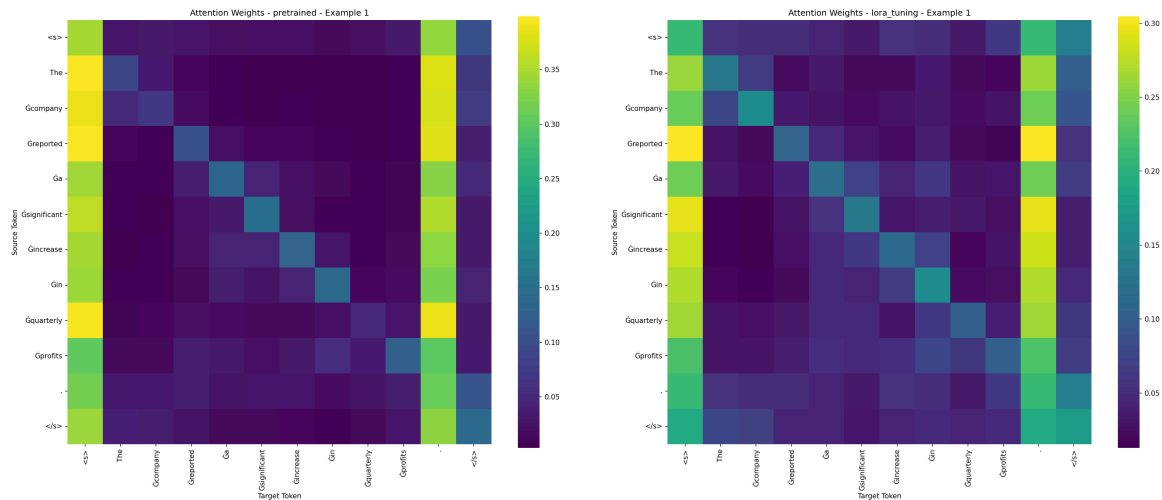


Figure 4: Pretrained RoBERTa (left) vs LoRA fine-tuned (right) - attention patterns

Observations: Fine-tuning shifts attention from function words to domain-specific semantic terms ("increase," "quarterly," "profits"), capturing financial term relationships.

4. Discussion

4.1 Performance-Efficiency Trade-off

LoRA's 2.59% accuracy drop is acceptable given 140× parameter reduction and 2.35× speedup. With optimal hyperparameters ($r=8$, $\alpha=64$), LoRA even surpasses full fine-tuning (80.31% vs 79.79%).

Recommendations: (1) Multiple tasks → LoRA (store many adapters); (2) Resource-constrained → LoRA (lower memory/storage); (3) Single high-stakes task → Full fine-tuning (maximum performance); (4) Rapid prototyping → LoRA (faster iteration).

4.2 Why LoRA Works

(1) **Low-rank hypothesis:** Task adaptations lie in low-dimensional subspaces; (2) **Knowledge preservation:** Frozen weights retain general understanding; (3) **Targeted updates:** Only modifies attention mechanisms; (4) **Regularization:** Low-rank constraint prevents overfitting.

4.3 Limitations

Small dataset (961 samples) may not generalize to larger scales; class imbalance affects neutral class performance; single base model tested (RoBERTa-base); no comparison with domain-specific models (e.g., FinBERT).

5. Conclusion

This study demonstrates that LoRA achieves 77.20% accuracy with 0.71% trainable parameters and 2.35× faster training, providing an excellent efficiency-performance trade-off. Both methods improve financial sentiment understanding by ~20 percentage points. Attention visualization reveals fine-tuning reshapes model focus toward domain-specific terms. LoRA is ideal for resource-constrained and multi-task scenarios, while full fine-tuning suits single high-stakes applications. As models grow larger, parameter-efficient methods like LoRA become essential for democratizing NLP capabilities.

Future Directions: Test on larger models (RoBERTa-large, GPT-3); compare other PEFT methods (Adapters, Prefix-tuning); multi-task learning with multiple LoRA adapters; theoretical analysis of low-rank sufficiency.

6. References

1. Liu, Y., et al. (2019). RoBERTa: A robustly optimized BERT pretraining approach. *arXiv:1907.11692*.
 2. Hu, E. J., et al. (2021). LoRA: Low-rank adaptation of large language models. *arXiv:2106.09685*.
 3. Vaswani, A., et al. (2017). Attention is all you need. *NeurIPS*.
 4. Devlin, J., et al. (2019). BERT: Pre-training of deep bidirectional transformers. *NAACL*.
 5. Hounsby, N., et al. (2019). Parameter-efficient transfer learning for NLP. *ICML*.
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Appendix: Core Code Snippets

Dataset Preprocessing:

```
class FinancialPhraseDataset(Dataset):
    def __init__(self, data_path, tokenizer, max_length=128):
        self.data = pd.read_csv(data_path)
        self.tokenizer = tokenizer
        self.max_length = max_length

    def __getitem__(self, idx):
        text = self.data.iloc[idx]["sentence"]
        label = self.data.iloc[idx]["label"]
        encoding = self.tokenizer(text, max_length=self.max_length,
                                   padding="max_length", truncation=True)
        return {"input_ids": encoding["input_ids"],
                "attention_mask": encoding["attention_mask"],
                "labels": torch.tensor(label)}
```

Evaluation:

```
def evaluate_model(model, dataloader, device):
    model.eval()
    predictions, true_labels = [], []
    with torch.no_grad():
        for batch in dataloader:
            batch = {k: v.to(device) for k, v in batch.items()}
            outputs = model(**batch)
            preds = torch.argmax(outputs.logits, dim=1)
            predictions.extend(preds.cpu().numpy())
            true_labels.extend(batch["labels"].cpu().numpy())
    return {"accuracy": accuracy_score(true_labels, predictions),
            "f1": f1_score(true_labels, predictions, average="weighted")}
```

Complete code: https://github.com/Yo1ogreyZz/DSA4213_Natural-Language-Processing-for-Data-Science.git